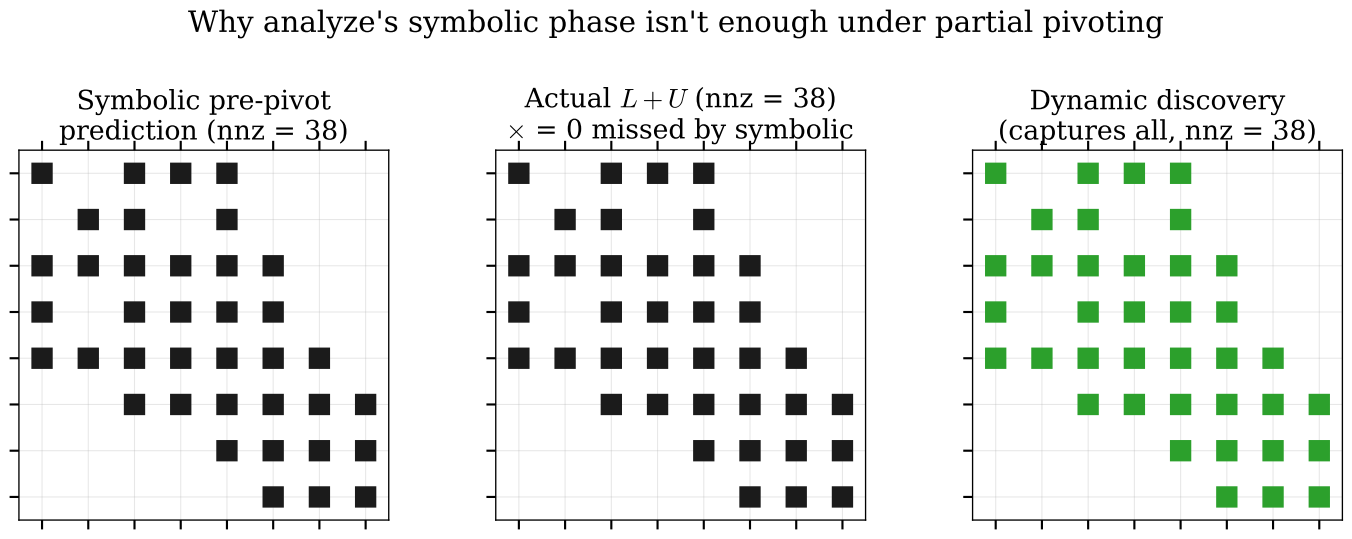

Why analyze's symbolic phase isn't enough under partial pivoting
Symbolic pre-pivot prediction (nnz = 38)
Actual $L+U$ (nnz = 38)
× = 0 missed by symbolic
Dynamic discovery (captures all, nnz = 38)